Shishir Naresh

Data Scientist
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Languages Known: *English, Hindi*

Objective:

*Data Scientist with **8+ years of experience** working on data-driven systems, machine learning models, and large-scale analytics platforms across telecom, security, and enterprise domains. Strong background in **feature engineering, statistical analysis, anomaly detection, causal inference, and model evaluation**, with hands-on experience deploying models into production. Proven expertise in Generative AI, **LLM-based applications, and Retrieval-Augmented Generation (RAG)**, combined with a solid foundation in **Big Data, cloud platforms (AWS, GCP),** and scalable data pipelines. Adept at converting complex data into meaningful insights that drive business and engineering outcomes.*

Summary:

- *Data scientist, AI/ML engineer with expertise in **Python, Machine Learning, Deep Learning, Big Data technologies, Analytics, and backend development**.*
- *Proven experience in managing projects, leading teams, and performing database migrations and architecture improvements.*
- ***Strong analytical, problem-solving, and good communication skills** with a track record of innovation and excellence in technical performance.*
- *Educational background includes a **Masters in AI and ML, Bachelor of Technology in Electrical Engineering** with high academic achievements.*
- ***Awarded multiple times for technical excellence, innovation**, and team performance in professional settings.*
- ***2+ Years of experience in semantic data modelling techniques** in Graph database (Neo4j) and Knowledge Graph development.*

Technical Skills:

Programming Languages: *Python, SQL, Node.js, Typescript*
Frameworks: *Flask, TensorFlow, PyTorch, Express.js, Fast Api*
Database: *PostgreSQL, Neo4j, MongoDB, MariaDB, MySQL*
Big Data: *Databricks, PySpark, Hadoop, Google Big Query*
Cloud: *AWS, Azure*
Version Control System and CI/CD: *GIT, Azure DevOps*
Integrated Development Environment: *Vs Code, PyCharm, IntelliJ Idea*
Data visualization: *Tableau*
Skills: *AI, ML (Machine Learning), NLP (Natural Language Processing), Deep Learning, NER(Named Entity Recognition), Anomaly Detection, Transfer Learning, BERT, Data Mining, Advanced Analytics, Data Collection, Data Preparation, Data Processing, Good communication*
Operating systems: *Mac, Linux, Windows*
Domain: *Life Sciences, Cybersecurity*

Education:

MTech in Artificial Intelligence and Machine Learning, Nov 2024
Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Technology: Electrical Engineering, May 2015
Uttarakhand Technical University, Dehradun
Percentage: 75.6
Class-XII: St Mary's Sr Sec School, 2011
CBSE, Haridwar
Percentage: 85.6
Class-X: St Mary's Sr Sec School, 2009
CBSE, Haridwar
Percentage: 90

Experience:

AI ML Engineer (November 2024-Present)
UST, Noida
Role: *AI ML Engineer*
Technologies: *Python, Pandas, GIT, Linux, AWS, PySpark, S3 Bucket, Step Function, CloudWatch, Fargate, Lambda, FAST API, Flask, Tensor flow, PyTorch, Graph Neural Networks, Causal Forest* ***Database****: MySQL, PostgreSQL, PySpark, SQL, Neo4j*

Client: Nokia – Network Optimization, Automation & GenAI

- ***GenAI LLM-Based Multi-Tenant Chatbot Platform***
  - *Designed and implemented an enterprise-scale Generative AI chatbot platform supporting multi-tenant and multi-chatbot architectures.*
  - *Built Retrieval-Augmented Generation (RAG) pipelines using Qdrant vector database, including embedding generation, chunking strategies, and similarity search.*
  - *Developed LangGraph-based agentic workflows for multi-step reasoning, tool invocation, and API orchestration.*
  - *Designed data modeling layers for structured and unstructured knowledge sources to improve retrieval accuracy and reduce hallucinations.*
  - *Integrated backend enterprise APIs and applied prompt engineering and response validation to enhance response quality.*
  - *Evaluated system performance using retrieval precision, response relevance, and latency metrics.*

- ***Telecom Call Failure Root Cause Analysis (Vox Project)***
  - *Developed a data analytics and RCA pipeline to analyze telecom call failures and voicemail drops using TAS, AMF, and MME logs.*
  - *Performed feature engineering and reconstructed end-to-end call lifecycles using unique call identifiers.*
  - *Conducted comparative statistical analysis between successful and failed calls to identify failure patterns.*
  - *Categorized calls into distinct RCA classes, enabling structured troubleshooting and reporting.*
  - *Delivered insights through EDA and visual analytics to support engineering and operations teams.*

- ***Vision AI – Pole Height Estimation (ATT AIO Project)***
    - *Designed a scalable Vision AI pipeline on Google Cloud Platform to extract pole height from 2D/3D street images.*
    - *Integrated Vertex AI Gemini Flash vision-language model for multimodal inference.*
    - *Built Cloud Run–based asynchronous batch pipelines using task queues and worker pools.*
    - *Applied prompt engineering to improve extraction accuracy and consistency.*
    - *Persisted structured outputs to Cloud Storage and BigQuery for downstream analytics.*

- ***Network Optimization – Advanced ML Modeling***
    - *Performed data cleaning and feature engineering for telecom fault alarm forecasting.*
    - *Built and evaluated ML models addressing class imbalance and noisy labels.*
    - *Modeled telecom data into knowledge graphs and trained models for anomaly detection and event prediction.*
    - *Implemented Causal Double Machine Learning (DML) to identify parameters with true causal impact on spectral efficiency.*
    - *Applied entropy, mutual information, and hypothesis testing to validate feature relevance and prevent overfitting.*

- ***Fault Alarm Forecasting:***

    - *Perform data cleaning and feature engineering, to understand the noise and patterns in the data.*
    - *Developed different modelling techniques around the data, so that model can learn easily from it.*
    - *Improved model performance by mitigating different issues and addressing class imbalance issues.*
    - *Model data into knowledge graphs and train models on top of them for anomaly detection and event prediction.*

- ***Spectral Efficiency Parameter optimization:***

    - *Due to a large number of parameters, implemented data cleaning and feature engineering techniques to include only those features which are having good variances around the features and having good impact on the target variable.*
    - *Implemented Causal Double ML model to understand the features that are impacting the target variables.*
    - *As there is non-linearity between the features, so understand the results better and implement different metrics to check whether the model is overfitting or underfitting.*
    - *Performed hypothesis testing using entropy and mutual information around the features.*
    - *As due to introduction of different KPI to the features, it was impacting the performance of the model. It was becoming overfit due to high correlation between target variable and KPI features. So addressed this issue by reducing the dimension of the data.*

***Senior Software Engineer* (March 2022- Nov 2024)**
Accenture, Noida
Role: *Application developer*
Technologies: *Python, Pandas, Node.js, Typescript, GIT, Linux, AWS, MySQL, PySpark, Google Big Query, S3 Bucket, Step Function, CloudWatch, Fargate, Lambda, FAST API, Flask Database: MySQL, PostgreSQL, Big Query, PySpark, DynamoDB, SQL*

Information Security Dashboard

- *Develop backend ingestion logic for different ETL and consume or upload files into S3 bucket.*
- *Migrate the whole project from MySQL to Postgres SQL.*
- *Perform POCs and provide estimates for new RFCs*
- *Perform architecture and DB improvements, so the application performance can be improved.*
- *Migration of Database from Postgres SQL to Google Big Query and change application architecture such that Backend works with Postgres and Frontend talks with Google Big Query and both databases remain in sync in real time using Kafka.*
- *Maintain and modify existing ETL code.*
- *Solve and deploy multiple new functionalities and bugs using pre-written pipelines.*
- *Work on new functionalities and design them into existing architecture without affecting existing functionalities.*

***Software Engineer*, (Jan 2016 – March 2022)**
Tata Consultancy Services, Noida
Role: *Full Stack Developer*
Technologies: *Java, Spring Boot, MVC, Node.js, Python, TypeScript, GIT, Linux, Neo4j, Bot Framework, JWT*
Database*: Neo4j, MariaDB, MongoDB*

TCS Life Sciences Client Project (May'19- Present)

- *Part of TCS Advisory development team for creating genomics platform.*
- *Developed semantic knowledge graph for clinical data and published the white paper on the same.*
- *Implemented multiple AI and ML solutions using python and exposed them to the application in the form of REST Api and implemented a security layer on top of it.*
- *Implemented logging scheme for the application and performed data modeling using ORM in java.*
- *Implemented ingestion logic for Knowledge graph creation in neo4j for the platform.*
- *Implemented UI for showing the data in the form of a knowledge graph.*

TCS Cloud Business Unit ITSM Client Project (Feb'18- Apr'19)

- *Was part of the development team for creating a ITSM chatbot platform.*
- *Implemented multiple REST Apis and managed the service now for ticketing by chatbot.*
- *Implemented user session management for the bot timeout using JWT.*
- *Implemented data modelling for the data using TypeORM in node.js.*
- *Integrated multiple new AI and ML cloud-based solutions in the chatbot. For example, an integrated ImagetoText Cloud based solution for users to upload error images and tickets can be created for the same.*
- *Integrated Cortana voice channel with the chatbot. So that user can interact with bot based on voice search.*
- *Managed Microsoft azure cloud infra for the application deployment on the cloud and integrated multiple channels for the bot for e.g. Direct line, MS Team and skype.*

TCS Lifesciences Client Project (Apr'16- Jan'18)

- *Was part of the development team for digitalization of the clinical trial process.*
- *Implemented multiple REST Apis and performed code modification on the whole codebase to reduce the execution time.*
- *Took initiative to synchronize the data from multiple sources and was accessible at a single place for analysis.*
- *Implemented upload and buffering REST Api of multimedia data into GridFs MongoDB store in the form of chunks, so it can be streamed for any new user of the application.*

ILP Project (Jan'16- Apr'16)

- *Worked on this project as a part of TCS ILP (Initial Learning Program).*
- *Was responsible for handling media files in the application which involves uploading the file linked to a particular schedule.*
- *Led the team for this project.*
- *Was also awarded the ILP **LIREL (**Awarded to the best associate during training**)***

Awards:

- *Innovation Pride Award, Award for Innovation, (10-Feb-2021)*
- *Service & Commitment Award (18-Jan-2021)*
- *Technical Excellence Award, Awards for Excellence(27-Aug-2020)*
- *On the Spot (Team) Award, Awards for Excellence (18-Aug-2020)*
- *Applause for Team Award, Beyond Performance Awards (07-Aug-2020)*
- *On The Spot Award, Awards for Excellence (June 2017)*
- *ILP LIREL (Awarded to the best associate during training), Tata Consultancy Services (July 2016)*

Extra-Curricular:

- *Active participation in TCS Maitree, TCS FitforLife and TCS Purposeforlife events.*
- *Active member of TCS Fire Committee.*
- *Highly enthusiastic and active participation in Technical/Non-Technical Events.*